Hierarchical Predictive Processing

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Neural Network Fundamentals

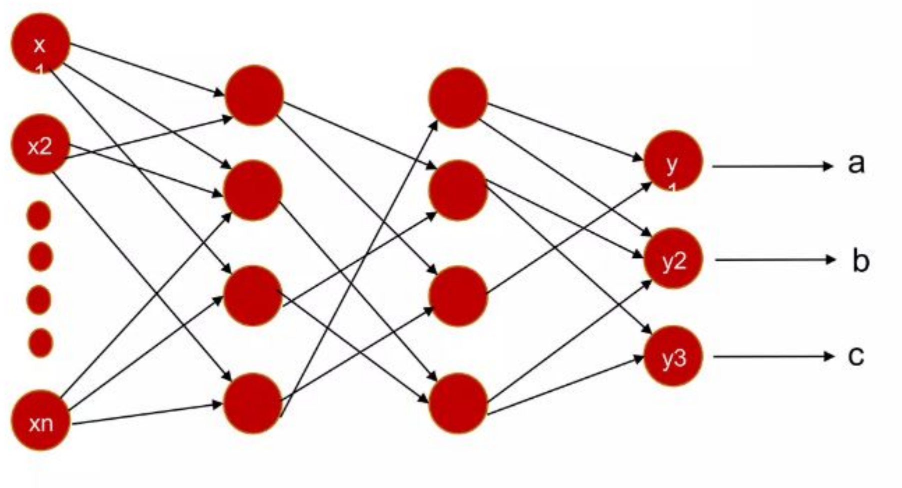
Core Objective

This simple neural network is specialized to recognize and classify the handwritten letters **'a'**, **'b'**, and **'c'**.

How it Works

The architecture processes input signals through hidden layers to determine the final probability of each character.

This process translates complex visual patterns into discrete categorical outputs.



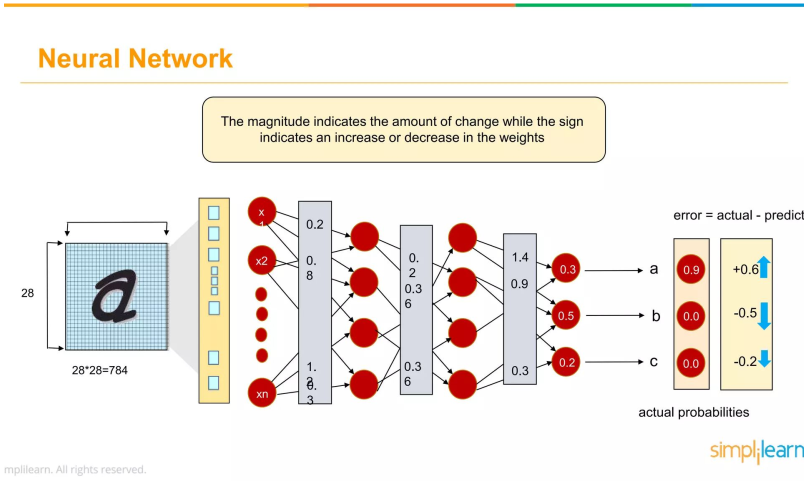
Forward Propagation

Input Layer

For instance, let's consider an image with dimensions of 28 x 28 pixels. In total, we have 784 pixels or neurons serving as input.

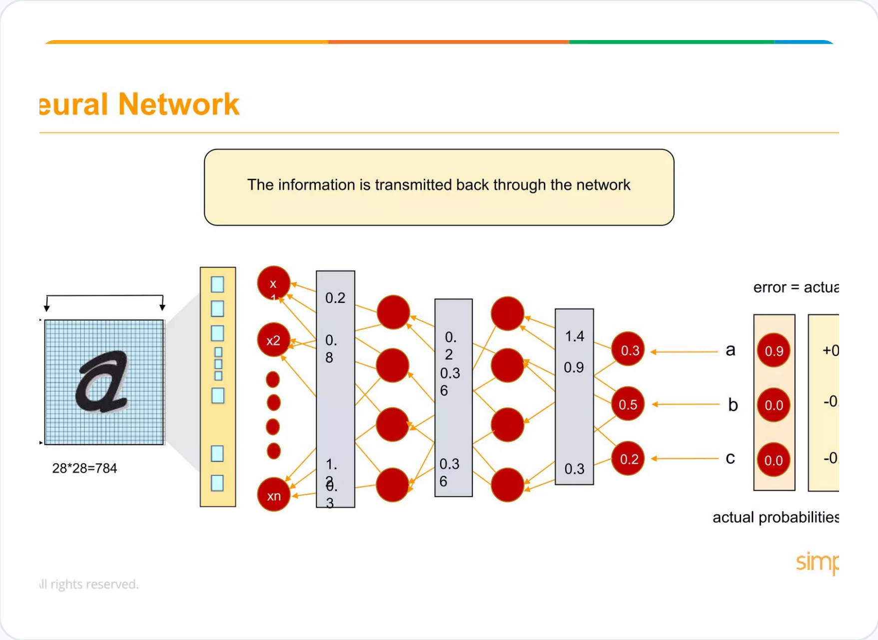
Weight Initialization

Initial predictions are generated through random weights, marking the beginning of forward propagation.



The predicted probabilities are compared against the actual probabilities and the error is calculated

Backpropagation

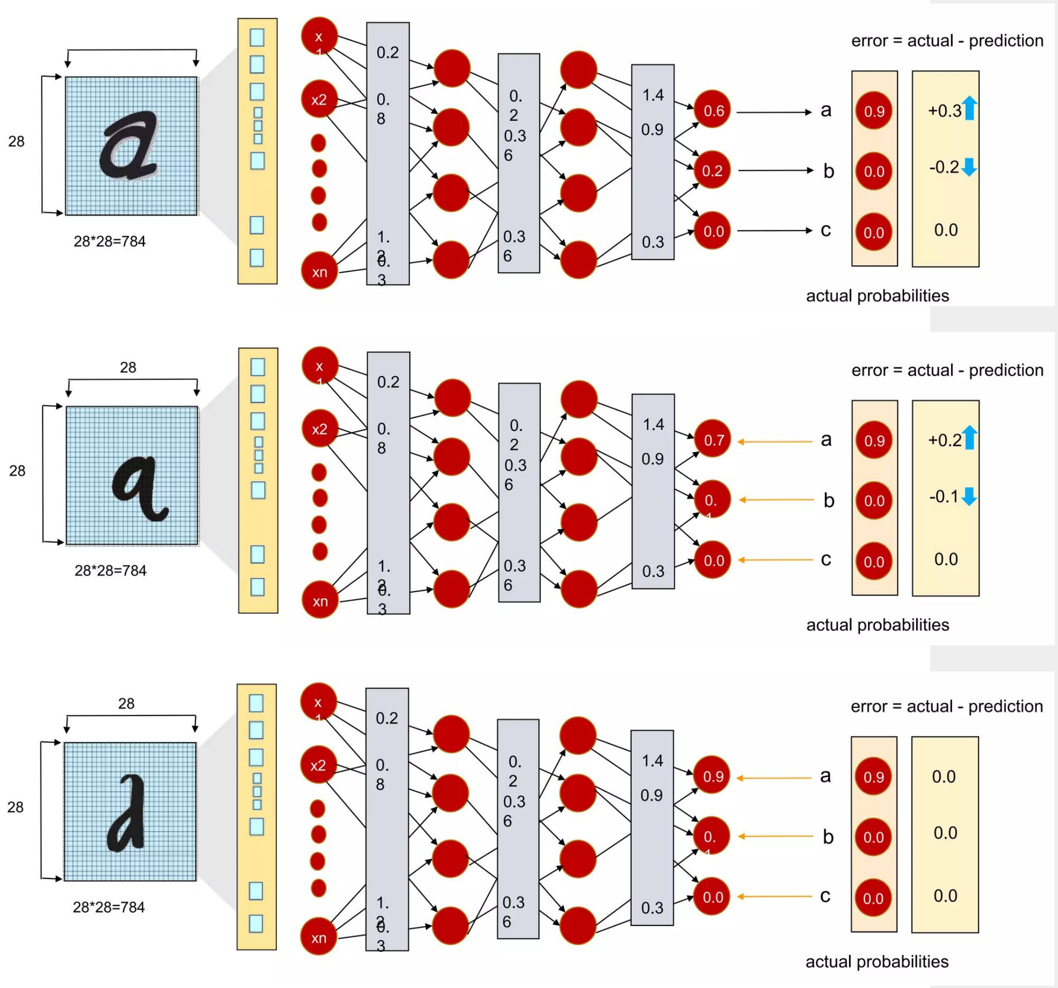


The Learning Mechanism

During training, the network compares predictions with actual outputs to calculate errors.

These errors are used to adjust connection weights between neurons, symbolized by blue flashes.

This iterative feedback process allows neural networks to learn from mistakes and improve accuracy.



Adjusting Weights

Weights throughout the network are continually adjusted to minimize prediction loss.

This iterative process involves training the network with multiple inputs until it achieves high prediction accuracy.

But there is no supervisor in the brain!

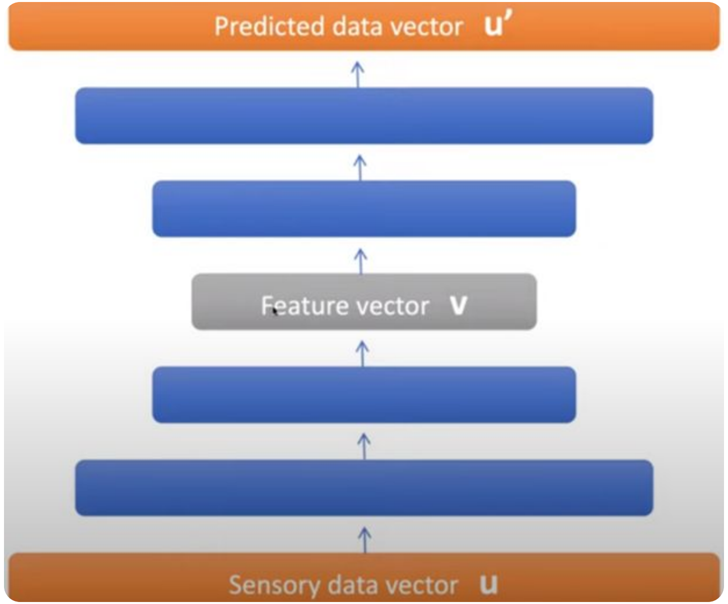
Unsupervised Learning

In the real world, data often lacks explicit labels like "cat" or "dog." Our brains must learn to derive patterns from raw data alone.

Unsupervised algorithms enable the discovery of inherent structures and relationships without labeled examples.

How does the brain recognize patterns in the absence of a supervisor?

Predictive Simulation & Feature Learning



Predictive Coding in the Brain

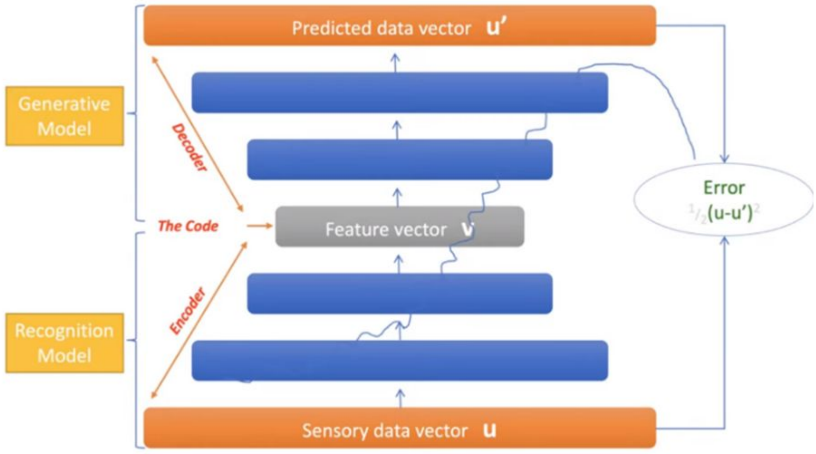
Through observation, we form **assumptions** about the world to anticipate perceptions.

By simulating assumed features, our brains predict what we're likely to encounter.

Neural Network Mirroring

Autonomous **feature learning** enables networks to simulate and predict future observations, mimicking biological intelligence.

Predictive Error (PE) in Network Training



Key Concepts

Predictive Generation

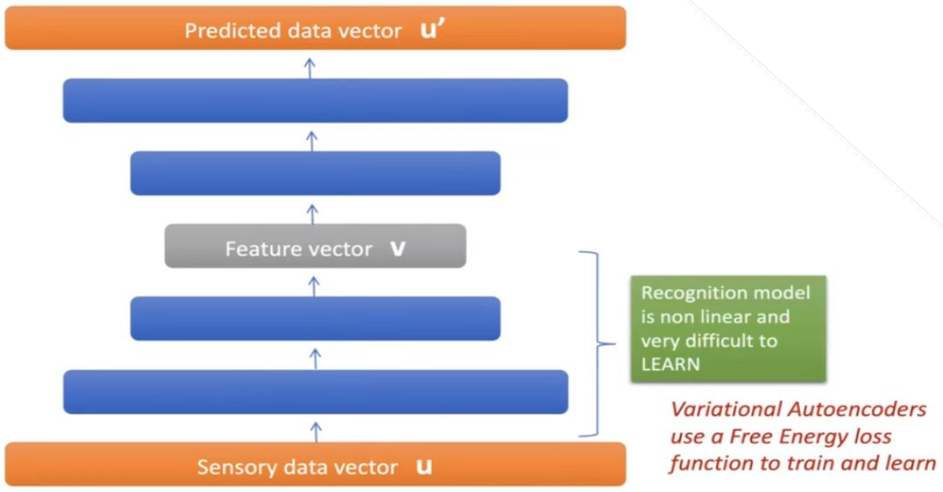
The generative model creates an internal simulation: *"is generating what I would be seeing"*

Learning Mechanism

The Feed Forward Model processes sensory inputs while PE (Prediction Error) is used to refine and train the neural network.

The brain minimizes the error (u - u') between sensory data and predictions.

Training Challenges in Recognition Models



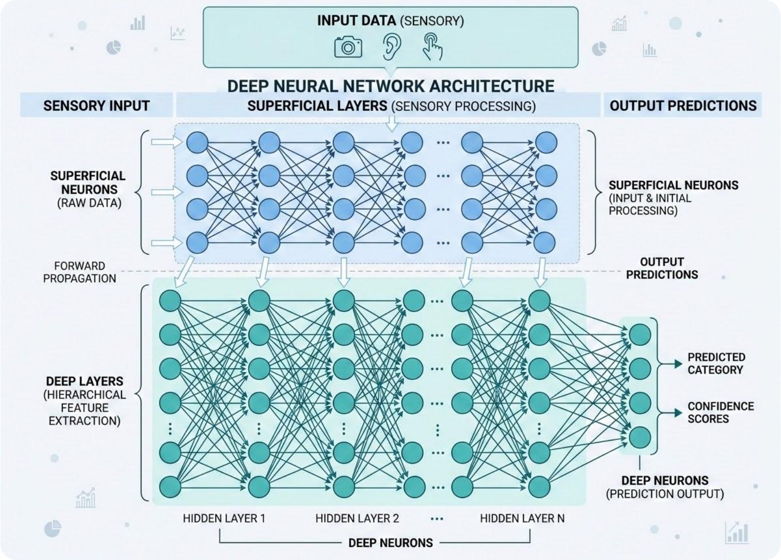
The Difficulty of Learning

Training this specific part of the network is exceptionally difficult due to several factors:

- Highly non-linear recognition models.
- Vast number of possibilities (search space).
- Ambiguous data that is difficult to decipher.

Variational Autoencoders utilize a Free Energy loss function to facilitate training and learning.

Brain Sensory Processing as Variational Autoencoders



Hierarchical Processing

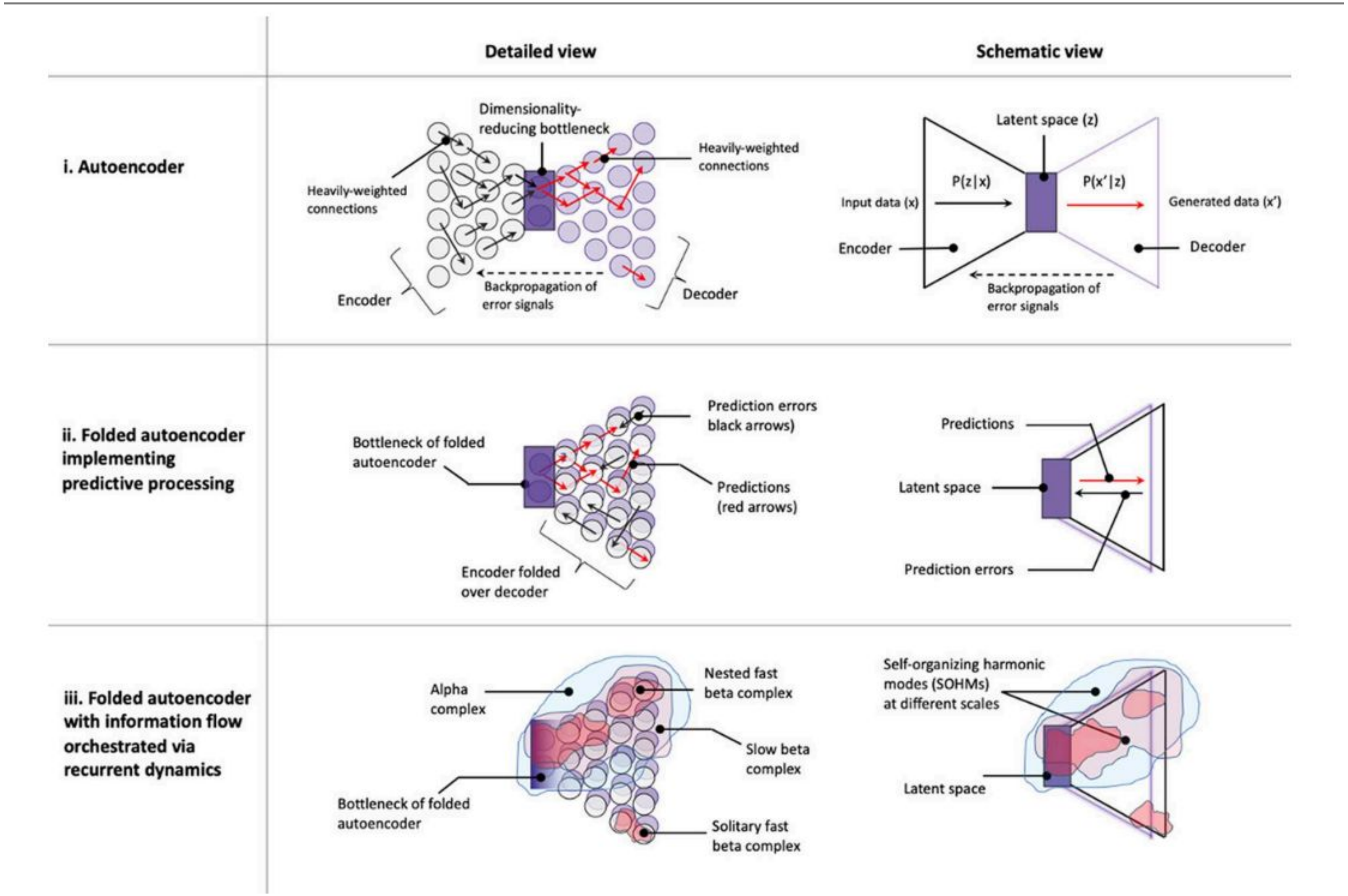
Neural Architecture

Encoders and decoders utilize distinct layers: superficial neurons handle sensory inputs, while deep neurons generate predictions.

HPP vs. Traditional Models

Unlike traditional autoencoders, HPP models minimize prediction errors across all levels simultaneously.

The brain constantly compares real-time sensory data with internal predictions to refine its environmental understanding.


Detailed view
Schematic view
i. Autoencoder
Dimensionality-reducing bottleneck
Heavily-weighted connections
Heavily-weighted connections
Encoder
Backpropagation of error signals
Decoder
Latent space (z)
P(z|x)
P(x'|z)
Input data (x)
Generated data (x')
Encoder
Decoder
Backpropagation of error signals
ii. Folded autoencoder implementing predictive processing
Bottleneck of folded autoencoder
Prediction errors black arrows)
Predictions (red arrows)
Encoder folded over decoder
Predictions
Latent space
Prediction errors
iii. Folded autoencoder with information flow orchestrated via recurrent dynamics
Alpha complex
Nested fast beta complex
Slow beta complex
Bottleneck of folded autoencoder
Solitary fast beta complex
Self-organizing harmonic modes (SOHMs) at different scales
Latent space

Variational autoencoder features	Proposed correspondences in predictive processing
Encoder network	Ascending hierarchy of superficial pyramidal neurons; Message-passing at gamma frequencies
Generative decoder network	Descending hierarchy of deep pyramidal neurons; Beliefs propagated at beta frequencies
Reduced dimensionality bottleneck	Association cortices and deeper portions of generative models; Estimates calculated at beta, alpha, and theta frequencies
Mean vectors	Activity levels for neuronal populations at different parts of hierarchy
Variance vectors	Neuronal population activity variability
Sampling from latent feature space	Large-scale synchronous complexes at beta, alpha, and theta frequencies; "ignition" events
Training: minimizing reconstruction loss between input layer of encoder and output layer of generative decoder; also minimizing divergence from unit Gaussian, parameterized by disentangling parameter	Training: minimizing precision-weighted prediction-errors at all layers simultaneously; precision-weighting as analogous to disentanglement hyperparameter; many mechanisms including synchronous gain control and diffuse neuromodulatory systems
Potential for sequential organization via recurrent network controllers (Ha and Schmidhuber, 2018)	Organization of state transitions by hippocampal system and frontal cortices (Koster et al., 2018)

Hierarchical Predictive Processing (HPP) in Sensory Cortices

Neural Architecture & Learning

HPP models approximate disentangled variational autoencoders, utilizing hierarchies of superficial and deep pyramidal neurons as encoders and decoders.

Prediction Error Minimization

Rather than training on input/output divergences, error is minimized at all levels simultaneously through real-time comparison of sensory data and internal predictions.

Advanced Cortical Integration

Brain function involves multiple interacting autoencoding hierarchies, with exceptionally strong connections in deeper association cortices.

- **Latent Spaces:** Corresponds to reduced dimensionality for efficient processing.
- **Synergistic Power:** Leverages shared priors from multi-modal integration.
- **World Modeling:** Facilitates comprehensive internal representations.

References

Simplilearn Tutorial

https://www.slideshare.net/Simplilearn/backpropagation-and-gradient-descent-in-neural-networks-neural-network-tutorial-simplilearn

Video Resource

https://www.youtube.com/watch?v=UkH-7gZnrr4&t=399s